Getting Started With Open-Source Verification

Setup and Prerequisites

- Demos are designed to be fully Open-Source
- They can be run on any mid-level PC

Item	Value
OS Name	Microsoft Windows 11 Home
Version	10.0.26100 Build 26100
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	
System Manufacturer	HP
System Model	HP Pavilion Plus Laptop 14-eh0xxx
System Type	
System SKU	
Processor	12th Gen Intel(R) Core(TM) i5-1240P, 1700 Mhz, 12 Core(s), 16 Logical Processor(s)
BIOS Version/Date	Insyde F.10, 16/08/2023
SMBIOS Version	3.3
Embedded Controller Version	31.36
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	8A36
BaseBoard Version	31.36
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United Kingdom
Hardware Abstraction Layer	Version = "10.0.26100.1"
Username	
Time Zone	GMT Summer Time
Installed Physical Memory (RAM)	8.00 GB

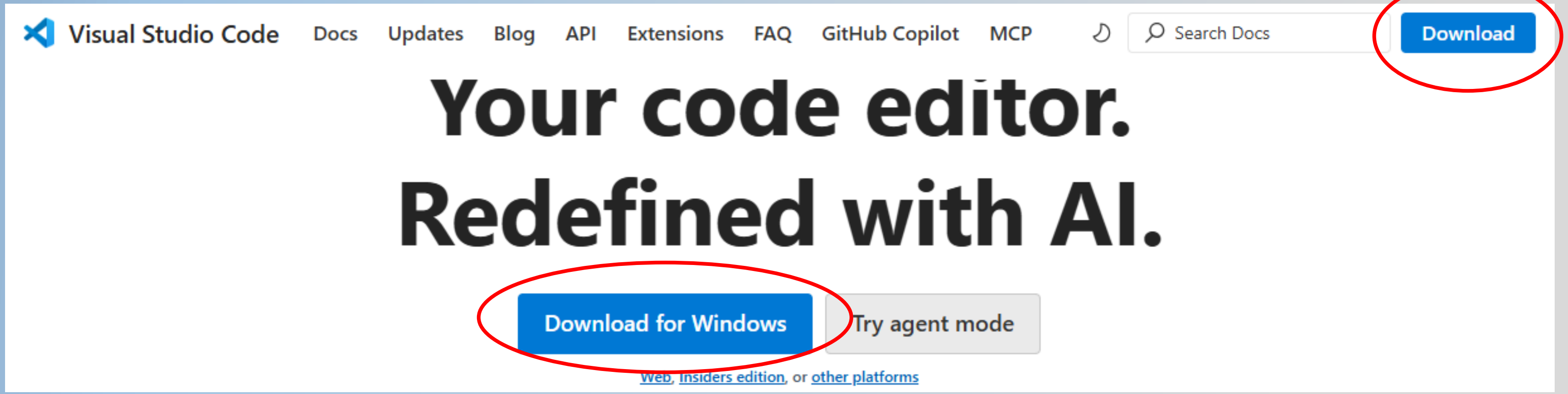
Setup

- This tutorial has been developed on Ubuntu running on a windows laptop under [WSL]

PowerShell   Copy

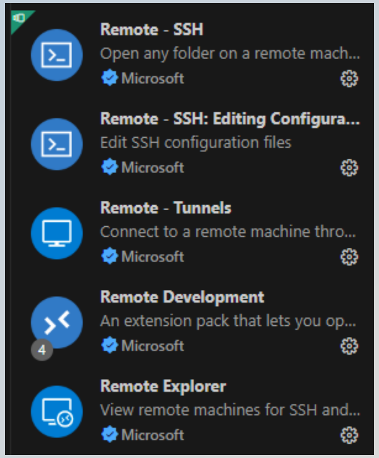
```
wsl --install
```

- The demos are run under [VSCode]



Setup

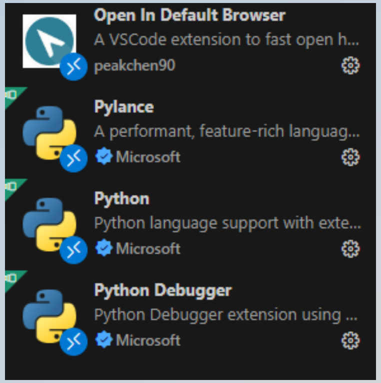
- Using VSCode with WSL
  - https://code.visualstudio.com/docs/setup/windows
  - https://code.visualstudio.com/docs/remote/wsl

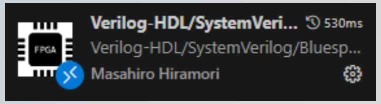


- Using VSCode on MacOS
  - https://code.visualstudio.com/docs/setup/mac

Setup

- For a better editor experience Python, SystemVerilog and Browser extensions are added to VSCode





Setup

- All examples are run using [Verilator]
- AVL requires a new version installed directly from Git

Git Quick Install

Installing Verilator from Git provides the most flexibility; for additional options and details, see Detailed Build Instructions below.

In brief, to install from git:

```
# Prerequisites:
#sudo apt-get install git help2man perl python3 make autoconf g++ flex bison ccache
#sudo apt-get install libgoogle-perftools-dev numactl perl-doc
#sudo apt-get install libfl2  # Ubuntu only (ignore if gives error)
#sudo apt-get install libfl-dev  # Ubuntu only (ignore if gives error)
#sudo apt-get install zlibc zlib1g zlib1g-dev  # Ubuntu only (ignore if gives error)

git clone https://github.com/verilator/verilator   # Only first time

# Every time you need to build:
unsetenv VERILATOR_ROOT  # For csh; ignore error if on bash
unset VERILATOR_ROOT  # For bash
cd verilator
git pull         # Make sure git repository is up-to-date
git tag          # See what versions exist
#git checkout master      # Use development branch (e.g. recent bug fixes)
#git checkout stable      # Use most recent stable release
#git checkout v{version}  # Switch to specified release version

autoconf         # Create ./configure script
./configure      # Configure and create Makefile
make -j `nproc`  # Build Verilator itself (if error, try just 'make')
sudo make install
```

Setup

- Waveforms can be viewed using GTKWave

GTKWave

GTKWave is a fully featured GTK+ based wave viewer for Unix and Win32 which reads FST, and GHW files as well as standard Verilog VCD/EVCD files and allows their viewing.

Building GTKWave from source

Installing dependencies

Debian, Ubuntu:

```
apt install build-essential meson gperf flex desktop-file-utils libgtk-3-dev \
            libbz2-dev libjudy-dev libgirepository1.0-dev
```

Fedora:

```
dnf install meson gperf flex glib2-devel gcc gcc-c++ gtk3-devel \
            gobject-introspection-devel desktop-file-utils tcl
```

macOS:

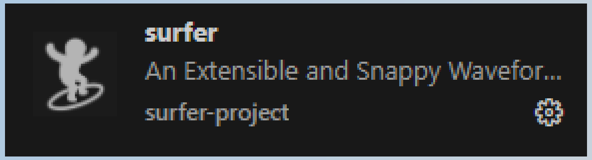
```
brew install desktop-file-utils shared-mime-info       \
             gobject-introspection gtk-mac-integration \
             meson ninja pkg-config gtk+3 gtk4
```

Building GTKWave

```
git clone "https://github.com/gtkwave/gtkwave.git"
cd gtkwave
meson setup build && cd build && meson install
```

Setup

- However, Surfer is a preferred Waveform viewer with VSCode integration



Setup

- Testbenches are designed using Cocotb

Setup

- AVL is on GitHub
  - https://github.com/projectapheleia/avl
  - https://github.com/projectapheleia/avl-apb
  - https://github.com/projectapheleia/avl-axi
  - https://github.com/projectapheleia/avl-axi-stream
- Full AVL documentation is available on ReadTheDocs
  - https://avl-core.readthedocs.io/en/latest/index.html

Setup

- AVL Tutorials are on GitHub
    - https://github.com/projectapheleia/avl-tutorials
    - This will install all other AVL libraries required


READY
TO
GO!
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